SOFTWARE REQUIREMENTS SPECIFICATION

RETINASCAN

A Deep Learning-Based Retinal Fundus Image Screening Tool for Early Detection of Diabetic Retinopathy

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List of Abbreviations

Term	Definition
DR	Diabetic Retinopathy
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
CLAHE	Contrast Limited Adaptive Histogram Equalization
Grad-CAM	Gradient-weighted Class Activation Mapping
IDRiD	Indian Diabetic Retinopathy Image Dataset
GPU	Graphics Processing Unit
EHR	Electronic Health Record
MSAG	Multi-Scale Spatial Attention Gating

1. Introduction

1.1 Purpose

Retina-Scan is a deep learning-based retinal fundus image screening system developed for the early pre-screening of Diabetic Retinopathy (DR). The system analyses retinal fundus images and classifies them into five severity levels: No DR, Mild DR, Moderate DR, Severe DR, and Proliferative DR.

The system uses EfficientNetB1 transfer learning implemented using PyTorch for classification, CLAHE for image enhancement, and Grad-CAM to provide a visual explanation of the prediction. The system is intended to support screening and referral decisions and does not replace professional medical diagnosis.

1.2 Scope

The scope of Retina-Scan covers the automated analysis of retinal fundus images through a web-based application. The image is uploaded by the user, preprocessed, classified using the trained EfficientNetB1 model, and presented with a predicted DR severity and confidence score.

The system also generates a Grad-CAM heatmap to indicate the regions of the retinal image that influenced the model prediction. The project uses the APTOS 2019 and IDRiD datasets for model development and evaluation.

1.3 Objectives

The main objectives of Retina-Scan are:

- To provide automated pre-screening of Diabetic Retinopathy.
- To classify retinal images into five DR severity levels.
- To use EfficientNetB1 with PyTorch for deep learning-based classification.
- To enhance retinal images using CLAHE.
- To handle class imbalance using class-weighted training.
- To provide prediction confidence.
- To generate Grad-CAM-based visual explanations.
- To provide a simple web interface for retinal image screening.
- To support early screening and referral prioritization.

1.4 Intended Audience

This SRS is intended for the project development team, project guide, evaluators, testers, and future developers who may maintain or extend the Retina-Scan system.

2. Overall Description

2.1 Product Perspective

Retina-Scan is a web-based deep learning system for pre-screening Diabetic Retinopathy from retinal fundus images. The system consists of a web interface, Flask backend, image preprocessing pipeline, EfficientNetB1 classification model, confidence estimation component, and Grad-CAM explainability module. The frontend allows users to upload retinal images and view the screening results, while the backend manages image processing, model inference, and result generation.

2.2 Product Functions

Major functions include retinal image upload and validation, image enhancement using CLAHE, image resizing and normalization, five-class Diabetic Retinopathy classification using EfficientNetB1, prediction confidence generation, Grad-CAM heatmap generation, and display of the severity result through the web interface. The system is intended to support early screening and referral prioritization.

2.3 User Classes

The primary user is a technician or authorized user who uploads retinal fundus images and views the screening results. Healthcare professionals can review the generated severity, confidence score, and Grad-CAM visualization to support further examination or referral.

2.4 Operating Environment

The application is intended to run as a responsive web application through a modern web browser. The frontend uses HTML, CSS, and JavaScript, while the backend is implemented using Flask. The deep learning component uses PyTorch and torchvision, with OpenCV for image processing and Grad-CAM for explainability. Model development and training can use GPU-supported environments with CUDA.

2.5 Assumptions and Dependencies

The system assumes that users provide suitable retinal fundus images with sufficient image quality for analysis. Prediction quality depends on the trained model and the datasets used for development and evaluation, including APTOS 2019 and IDRiD. The system also depends on the availability of the trained EfficientNetB1 model and required software libraries. The generated result is intended for pre-screening and referral support and must not be considered a definitive medical diagnosis.

3. System Analysis

3.1 Existing System

Existing Diabetic Retinopathy screening mainly relies on manual examination of retinal fundus images by trained ophthalmologists or specialists. Although automated approaches can assist with retinal image analysis, screening may still depend on specialist availability and manual interpretation. This can make large-scale and timely screening difficult, particularly in areas with limited access to eye-care professionals.

3.2 Limitations of Existing System

The main limitations are:

- Dependence on trained specialists for image examination.
- Time-consuming manual analysis of retinal images.
- Limited accessibility in areas with fewer ophthalmologists.
- Difficulty in conducting large-scale screening efficiently.
- Limited automated support for explaining classification decisions.
- Increased screening workload when large numbers of retinal images need to be examined.

3.3 Problem Statement

There is a need for an automated and accessible Diabetic Retinopathy pre-screening system that can analyse retinal fundus images, classify the severity of DR, and provide an understandable explanation of the prediction. Retina-Scan addresses this need by combining EfficientNetB1-based classification implemented using PyTorch, image preprocessing, confidence estimation, and Grad-CAM visualization in a web-based system.

3.4 Need for the Proposed System

The proposed system is needed to support faster and more accessible DR pre-screening. By automatically analysing retinal images and providing a severity prediction with a visual explanation, Retina-Scan can assist in identifying cases that may require further professional examination or referral.

4. Proposed System

4.1 Proposed Solution

Retina-Scan is a web-based deep learning system for Diabetic Retinopathy pre-screening using retinal fundus images. The system accepts a retinal image from the user and processes it through an image-processing and deep learning pipeline.

The image is enhanced using CLAHE, resized and normalized, and then classified using a trained EfficientNetB1 model implemented with PyTorch. The system classifies the retinal image into five severity levels: No DR, Mild, Moderate, Severe, and Proliferative DR.

The system also provides a prediction confidence score and generates a Grad-CAM heatmap to show the retinal regions that influenced the model's prediction. The result is displayed through the web interface to support pre-screening and referral.

4.2 Advantages

The proposed Retina-Scan system provides the following advantages:

- Automated pre-screening of Diabetic Retinopathy.
- Five-level DR classification using EfficientNetB1.
- PyTorch-based deep learning implementation.
- Image enhancement using CLAHE.
- Confidence score for the predicted class.
- Explainable prediction using Grad-CAM heatmaps.
- Web-based access through a simple user interface.
- Faster preliminary assessment of retinal images.
- Referral support for cases requiring further professional examination.

4.3 System Workflow

The Retina-Scan workflow consists of the following steps:

Retinal Fundus Image Upload → Image Preprocessing → EfficientNetB1 Classification → DR Severity Prediction → Confidence Score → Grad-CAM Generation → Screening Result

A typical Retina-Scan workflow begins when the user uploads a retinal fundus image through the web interface. The system validates the image and applies preprocessing techniques such as CLAHE enhancement, resizing, and normalization. The processed image is then passed to the EfficientNetB1 classifier, which predicts one of the five Diabetic Retinopathy severity levels.

The prediction module generates the confidence score, while the Grad-CAM module produces a heatmap highlighting the retinal regions that influenced the prediction. The final severity grade, confidence score, and Grad-CAM visualization are then displayed through the web interface for screening and referral support.

5. System Architecture

5.1 Architecture Overview

Retina-Scan follows a web-based deep learning architecture consisting of the user interface, Flask backend, image preprocessing pipeline, EfficientNetB1 classification model with Multi-Scale Spatial Attention Gating (MSAG), prediction and confidence module, and Grad-CAM explainability module. The user uploads a retinal fundus image through the web interface, while the backend manages image processing, model inference, and result generation.

5.2 Major Components

The major components of the system are:

- Web Interface: Allows the user to upload retinal images and view results.
- Flask Backend: Handles requests, image processing, model inference, and result delivery.
- Image Preprocessing: Enhances and prepares retinal images for classification.
- EfficientNetB1 Model: Classifies the retinal image into five DR severity levels.
- MSAG Module: Applies multi-scale spatial attention to the extracted feature representation.
- Prediction and Confidence Module: Generates the predicted class and confidence score.
- Grad-CAM Module: Generates a heatmap explaining the model prediction.
- Result Display: Presents the severity, confidence, and heatmap to the user.

5.3 Image Processing Pipeline

The uploaded retinal image passes through a preprocessing pipeline before classification. CLAHE is applied to improve image contrast and visibility, followed by image resizing and normalization to produce the required model input.

Input Image → CLAHE Enhancement → Resizing → Normalization → Model Input

5.4 EfficientNetB1 Classification

The processed retinal image is provided to the trained EfficientNetB1 model using transfer learning with PyTorch and torchvision. The extracted feature representation is processed using the Multi-Scale Spatial Attention Gating (MSAG) module before classification. The model classifies the image into five DR severity categories:

No DR → Mild → Moderate → Severe → Proliferative DR

Class-weighted training is used to reduce the effect of class imbalance during model training.

5.5 Grad-CAM Explainability

Grad-CAM is used to provide a visual explanation of the model's prediction. It generates a heatmap highlighting the regions of the retinal image that contributed to the predicted class. This helps make the automated prediction more interpretable.

5.6 Web Application Architecture

The web application consists of a frontend and Flask backend. The frontend handles image upload and result presentation. The Flask backend receives the image, performs preprocessing, invokes the PyTorch-based EfficientNetB1 model, generates the prediction and Grad-CAM output, and returns the results to the frontend.

User → Web Interface → Flask Backend → Preprocessing → EfficientNetB1 + MSAG → Prediction + Grad-CAM → Web Interface → User

6. Detailed Module Description

6.1 Image Upload Module

The Image Upload Module allows the user to upload a retinal fundus image through the web interface. It validates the image format and forwards valid images to the preprocessing module. Invalid or unsupported files are rejected with an appropriate error message.

6.2 Image Preprocessing Module

The Image Preprocessing Module prepares the uploaded image for classification. CLAHE is applied to enhance retinal image contrast, followed by resizing and normalization according to the EfficientNetB1 input requirements.

Input Image → CLAHE → Resizing → Normalization → Model Input

6.3 DR Classification Module

The DR Classification Module uses a trained EfficientNetB1 transfer learning model implemented using PyTorch to classify retinal images into five classes: No DR, Mild, Moderate, Severe, and Proliferative DR. The model incorporates the Multi-Scale Spatial Attention Gating (MSAG) module to enhance relevant spatial feature representations. The class with the highest predicted probability is selected as the final classification. Class-weighted training is used to reduce the effect of class imbalance.

6.4 Prediction and Confidence Module

This module processes the EfficientNetB1 output to determine the predicted DR severity and its corresponding confidence score. The highest probability class is selected as the prediction. The confidence value indicates the model's prediction strength and is not a medical diagnosis.

6.5 Grad-CAM Explainability Module

The Grad-CAM Module generates a heatmap showing the retinal regions that influenced the model's prediction. It uses feature maps and class-specific gradients from the EfficientNetB1 network to create an interpretable visualization.

Prediction → Gradients + Feature Maps → Grad-CAM Heatmap

6.6 Result Display Module

The Result Display Module presents the predicted DR severity, confidence score, original retinal image, and Grad-CAM visualization through the web interface. The output is intended for prescreening and referral support and should be reviewed by a qualified healthcare professional.

7. Functional Requirements

ID	Requirement
FR-01	The system shall allow users to upload retinal fundus images through the web interface.
FR-02	The system shall validate uploaded images and reject unsupported or invalid files.
FR-03	The system shall preprocess valid retinal images using CLAHE, resizing, and normalization.
FR-04	The system shall classify retinal images into five DR severity classes using the trained PyTorch-based EfficientNetB1 model.
FR-05	The system shall generate the predicted DR severity based on the highest classification probability.
FR-06	The system shall provide a confidence score corresponding to the predicted DR class.
FR-07	The system shall generate a Grad-CAM heatmap to explain the model prediction.
FR-08	The system shall display the predicted severity, confidence score, and Grad-CAM visualization through the web interface.
FR-09	The system shall handle image processing and prediction errors and provide appropriate user feedback.
FR-10	The system shall provide screening results for pre-screening and referral support and shall not represent them as a definitive medical diagnosis.

8. Non-Functional Requirements

ID	Requirement
NFR-01	Security: Uploaded retinal images and prediction data shall be protected against unauthorized access and processing.
NFR-02	Privacy: The system shall minimize the storage and exposure of user-uploaded retinal images and associated screening information.
NFR-03	Performance: The system should process a valid retinal image and provide the prediction and Grad-CAM result within a reasonable response time under normal operating conditions.
NFR-04	Usability: The web interface shall provide clear and understandable image upload and screening results for authorized users.
NFR-05	Accuracy: The system shall provide consistent classification results based on the trained EfficientNetB1 model and evaluation datasets.
NFR-06	Reliability: The system shall handle invalid images, processing errors, and model failures gracefully without causing application crashes.
NFR-07	Maintainability: The preprocessing, classification, Grad-CAM, and web components shall remain modular so that individual components can be updated independently.
NFR-08	Explainability: The system shall provide Grad-CAM visualizations to make model predictions more interpretable.
NFR-09	Compatibility: The web application shall operate on modern web browsers and supported Python, Flask, PyTorch, torchvision, and OpenCV environments.

9. External Interface Requirements

9.1 User Interface

The system shall provide a simple web-based interface for uploading retinal fundus images and viewing the screening results. The interface shall display the predicted DR severity, confidence score, original image, and Grad-CAM visualization.

9.2 Software Interface

The system shall use HTML, CSS, and JavaScript for the frontend and Flask for backend processing. PyTorch and torchvision shall be used for EfficientNetB1 model inference, while OpenCV shall support image preprocessing and Grad-CAM visualization.

9.3 Hardware Interface

The system shall operate on a computer capable of running the web application and deep learning model. GPU acceleration may be used during model development, training, and inference when a CUDA-compatible GPU is available, while inference may also be performed using CPU resources.

9.4 Model Interface

The trained EfficientNetB1 model shall receive a preprocessed retinal fundus image as input and produce probability values for the five DR severity classes. The prediction output shall be used to generate the final severity, confidence score, and Grad-CAM explanation.

10. Data Requirements

10.1 Input Data

The system shall accept retinal fundus images as input. The images shall be in a supported format and have sufficient quality for preprocessing and classification. The input image is processed using CLAHE, resizing, and normalization before being provided to the EfficientNetB1 model.

10.2 APTOS 2019 Dataset

The APTOS 2019 Diabetic Retinopathy Dataset is used for training and evaluation of the DR classification model. It contains retinal fundus images labelled according to five DR severity levels: No DR, Mild, Moderate, Severe, and Proliferative DR.

10.3 IDRiD Dataset

The Indian Diabetic Retinopathy Image Dataset (IDRiD) is used as an additional retinal image dataset for model development and evaluation. It provides labelled retinal fundus images and supports assessment of the model's ability to classify different DR severity levels.

10.4 Data Preprocessing

The retinal images shall undergo preprocessing before model training and inference. CLAHE is used to improve contrast, followed by resizing and normalization to produce consistent model inputs.

10.5 Data Augmentation

Data augmentation shall be applied during model training to improve generalization and reduce overfitting. Suitable transformations such as rotation, flipping, scaling, and brightness variation may be applied while preserving the relevant retinal features.

10.6 Class Imbalance Handling

Since the number of images may vary between DR severity classes, class-weighted training shall be used to reduce the effect of class imbalance and ensure that minority classes receive appropriate importance during model training.

11. Use Case Analysis

The Use Case Analysis describes the interaction between the user and the Retina-Scan system during the Diabetic Retinopathy pre-screening process.

11.1 Primary Use Case: DR Screening

Actor: Authorized User / Technician

Precondition: The Retina-Scan web application is available and the user has a valid retinal fundus image.

Main Flow:

- The user opens the Retina-Scan web application.
- The user uploads a retinal fundus image.
- The system validates the uploaded image.
- The system applies CLAHE, resizing, and normalization.
- The processed image is provided to the EfficientNetB1 model.
- The system predicts one of the five DR severity classes.
- The system calculates the prediction confidence.
- The Grad-CAM module generates a visual explanation.
- The system displays the severity, confidence score, and Grad-CAM visualization.

Alternative Flow: If the uploaded image is invalid or unsupported, the system displays an error message and requests a valid image.

11.2 Use Case Summary

Use Case	Actor	Description
Upload Image	User	Uploads a retinal fundus image.
Validate Image	System	Checks whether the image is valid and supported.
Preprocess Image	System	Applies CLAHE, resizing, and normalization.
Classify DR	System	Predicts one of five DR severity levels.
Generate Confidence	System	Provides the confidence score of the prediction.
Generate Explanation	System	Produces a Grad-CAM heatmap.
View Result	User	Views the severity, confidence, and visualization.

12. Hardware and Software Requirements

Category	Requirement
Frontend	HTML, CSS, JavaScript, and a modern web browser for the Retina-Scan interface.
Backend	Flask and Python for web services, image processing, and model inference.
Deep Learning	PyTorch, torchvision, and the trained EfficientNetB1 model with MSAG.
Image Processing	OpenCV for CLAHE enhancement, image preprocessing, and Grad-CAM generation.
Dataset Processing	Python libraries such as Pandas and NumPy for dataset preparation and analysis.
Development Environment	Visual Studio Code, Jupyter Notebook, Google Colab, or a suitable Python development environment for model development and testing.
Database/Storage	Local or server-side storage may be used for required application data and model files.
Hardware	A modern computer with sufficient RAM and storage to run the web application, image-processing operations, and deep learning model.
GPU	A CUDA-compatible GPU may be used to accelerate model training and inference; it is not mandatory for basic CPU inference.

13. Security and Privacy Requirements

The Retina-Scan system shall protect retinal images, prediction results, and other user-related information while ensuring secure access to the application.

- **Secure Image Handling:** Uploaded retinal images shall be processed securely and shall not be unnecessarily exposed or stored.
- **Access Control:** Access to uploaded images, prediction results, and system resources shall be restricted to authorized users.
- **Data Privacy:** Patient-related information, if collected, shall be kept confidential and used only for screening-related purposes.
- **Secure Communication:** Communication between the frontend and Flask backend shall use HTTPS/TLS in deployment environments.
- **Input Validation:** Uploaded files shall be validated for supported image formats and acceptable file properties before processing.
- **Malicious File Protection:** The system shall reject invalid or potentially malicious file uploads.
- **Model Protection:** The trained EfficientNetB1 model and associated configuration files shall be protected from unauthorized modification or replacement.
- **Result Integrity:** Prediction results and Grad-CAM outputs shall be generated from the processed input image and shall not be altered without authorization.
- **Temporary Data Protection:** Temporary uploaded images and generated heatmaps shall be securely handled and removed when they are no longer required.
- **Medical Privacy:** The system shall not disclose retinal images or screening results to unauthorized third parties.
- **Diagnostic Limitation:** The generated prediction shall be presented as a pre-screening result and shall not be represented as a definitive medical diagnosis.
- **Dataset Privacy:** APTOS 2019 and IDRiD dataset images shall be used according to their applicable usage and licensing conditions.

14. System Constraints

The Retina-Scan system is subject to the following constraints:

- **Image Quality:** Prediction accuracy depends on the quality, clarity, and suitability of the uploaded retinal fundus images.
- **Dataset Limitations:** Model performance is limited by the size, quality, class distribution, and representativeness of the APTOS 2019 and IDRiD datasets.
- **Model Accuracy:** The system may produce incorrect predictions, particularly for poor-quality images or cases that differ from the training data.
- **Computational Resources:** EfficientNetB1 inference and Grad-CAM generation require adequate computational resources.
- **Processing Time:** The response time may vary depending on image size, server performance, and available hardware.
- **Technology Dependency:** The system depends on Flask, PyTorch, torchvision, OpenCV, and other required software libraries.
- **Internet Dependency:** The web application may require network connectivity for communication between the user interface and backend when deployed remotely.
- **Medical Limitation:** The system is designed only for pre-screening and referral support and shall not be considered a replacement for professional ophthalmological diagnosis.

15. Testing and Evaluation Requirements

The Retina-Scan system shall be tested to ensure the correctness, reliability, performance, and usability of the complete screening workflow.

- **Unit Testing:** Individual modules such as image upload, preprocessing, classification, confidence calculation, and Grad-CAM generation shall be tested independently.
- **Integration Testing:** The interaction between the Flask backend, preprocessing pipeline, EfficientNetB1 model, prediction module, Grad-CAM module, and web interface shall be tested.
- **Model Evaluation:** The DR classification model shall be evaluated using accuracy, precision, recall, F1-score, and confusion matrix.
- **Class-wise Evaluation:** Performance shall be evaluated separately for No DR, Mild, Moderate, Severe, and Proliferative DR classes.
- **Image Processing Testing:** CLAHE enhancement, resizing, normalization, and image validation shall be tested using valid and invalid retinal images.
- **Prediction Testing:** The system shall be tested to verify that the predicted severity class and confidence score are generated correctly.
- **Explainability Testing:** Grad-CAM shall be tested to ensure that a valid heatmap is generated for supported predictions.
- **Web Application Testing:** Image upload, result display, error handling, and user interaction shall be tested across supported browsers.
- **Performance Testing:** The response time for image processing, model inference, and result generation shall be evaluated under normal operating conditions.
- **Security Testing:** File validation, unauthorized access, secure communication, and protection of uploaded images shall be tested.
- **Usability Testing:** The interface shall be evaluated for clarity, ease of image upload, readability of results, and understandability of the Grad-CAM visualization.
- **End-to-End Testing:** The complete workflow from image upload → preprocessing → classification → confidence calculation → Grad-CAM generation → result display shall be tested.

16. Future Scope

The future scope of Retina-Scan includes the following enhancements:

16.1 Cross-Population Validation

The model can be tested and fine-tuned using diverse population datasets, including IDRiD for Indian patients, to improve its generalization across different patient demographics and retinal imaging equipment.

16.2 Continuous Learning

The system can be periodically retrained using newly available labeled retinal images. This can help improve model performance and adaptability over time.

16.3 Integration with Hospital EHR Systems

Retina-Scan can be integrated with hospital Electronic Health Record (EHR) systems to connect screening results with patient records and support streamlined referral and follow-up.

16.4 Detection of Other Retinal Diseases

The existing deep learning pipeline can be extended to screen for other retinal diseases, such as Glaucoma and Age-Related Macular Degeneration (AMD), using suitable datasets and trained classification models.

17. Conclusion

Retina-Scan demonstrates the potential of deep learning and explainable artificial intelligence to improve the early screening of Diabetic Retinopathy, one of the leading causes of preventable blindness among diabetic patients.

By integrating PyTorch-based EfficientNetB1 transfer learning, Multi-Scale Spatial Attention Gating (MSAG), CLAHE-based image enhancement, class-weighted training, and Grad-CAM visualization, the system provides efficient and interpretable predictions from retinal fundus images. Its web-based design enables quick and accessible pre-screening, particularly in regions where access to ophthalmologists is limited.

Although Retina-Scan is intended to assist healthcare professionals rather than replace clinical diagnosis, it has the potential to reduce screening time, support timely referrals, and contribute to the early detection and treatment of diabetic eye disease. This can ultimately help preserve vision and improve patient outcomes.

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